

7

Free electrons

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In this chapter we investigate the optical properties associated with free electrons. As the name suggests, these are systems in which the electrons experience no restoring force from the medium when driven by the electric field of a light wave. The two main solid-state systems that exhibit strong free electron effects are:

- **Metals.** Metals contain large densities of free electrons that originate from the valence electrons of the metal atoms.
- **Doped semiconductors.** n-type semiconductors contain free electrons, while p-type materials contain free holes. The free carrier density is determined by the concentration of impurities used for the doping.

We begin our discussion of the optical properties by using the Drude–Lorentz model introduced in Section 2.1.3 of Chapter 2. This will enable us to explain the main optical property of metals that we mentioned in Section 1.4.3, namely that they reflect strongly in the visible spectral region. We then apply our knowledge of interband transitions from Chapter 3 to obtain a better understanding of the detailed form of the reflectivity spectra of metals such as aluminium and copper. Next we apply the Drude–Lorentz model to doped semiconductors to explain why doping causes infrared absorption. We then consider the collective oscillations of the whole free carrier gas, which will naturally lead us to the notion of plasmons, both bulk and surface. Finally, we conclude with a brief discussion of negative refraction.

7.1 Plasma reflectivity

A neutral gas of charged particles is called a **plasma**. Metals and doped semiconductors can be treated as plasmas because they contain equal numbers of fixed positive ions and free electrons. The free electrons experience no restoring forces when they interact with electromagnetic waves. This contrasts with bound electrons that have natural resonant frequencies in the near-infrared, visible, or ultraviolet spectral regions owing to the restoring forces of the medium.

In this section we derive a formula for the relative permittivity of an electron plasma using the classical oscillator model discussed in Section 2.2 of Chapter 2. As noted in Section 2.1.3, this approach combines the **Drude model** of free electron conductivity with the **Lorentz**

model of dipole oscillators, and is therefore known as the **Drude-Lorentz model**.

We begin by considering the oscillations of a free electron induced by the AC electric field $\mathcal{E}(t)$ of an electromagnetic wave. The equation of motion for the displacement x of the electron is:

$$m_0 \frac{d^2 x}{dt^2} + m_0 \gamma \frac{dx}{dt} = -e\mathcal{E}(t) = -e\mathcal{E}_0 e^{-i\omega t}, \quad (7.1)$$

where ω is the angular frequency of the light, and $\mathcal{E}_0$ is its amplitude. The first term represents the acceleration of the electron, while the second is the frictional damping force of the medium. The term on the right-hand side is the driving force exerted by the light. Equation 7.1 is the same as the equation of motion for a bound oscillator given in eqn 2.5, except that there is no restoring force term because we are dealing with free electrons.

By substituting $x = x_0 e^{-i\omega t}$ into eqn 7.1, we obtain

$$x = \frac{e\mathcal{E}}{m_0(\omega^2 + i\gamma\omega)}. \quad (7.2)$$

The polarization P of the gas is equal to $-Nex$, where N is the number of electrons per unit volume. By recalling the definitions of the electric displacement D and the relative permittivity ϵ_r (cf. eqns A.2 and A.3), we can write:

$$\begin{aligned} D &= \epsilon_r \epsilon_0 \mathcal{E} \\ &= \epsilon_0 \mathcal{E} + P \\ &= \epsilon_0 \mathcal{E} - \frac{Ne^2 \mathcal{E}}{m_0(\omega^2 + i\gamma\omega)}. \end{aligned} \quad (7.3)$$

Therefore:

$$\epsilon_r(\omega) = 1 - \frac{Ne^2}{\epsilon_0 m_0} \frac{1}{(\omega^2 + i\gamma\omega)}. \quad (7.4)$$

This equation is identical to eqn 2.14 for the bound oscillator except that the resonant frequency ω_0 is zero and we have not yet considered the effects of background polarizability. Equation 7.4 is frequently written in the more concise form:

$$\epsilon_r(\omega) = 1 - \frac{\omega_p^2}{(\omega^2 + i\gamma\omega)}, \quad (7.5)$$

where

$$\omega_p = \left(\frac{Ne^2}{\epsilon_0 m_0} \right)^{1/2}. \quad (7.6)$$

ω_p is known as the **plasma frequency**.

Let us first consider a lightly damped system. In this case, we put $\gamma = 0$ in eqn 7.5 so that

$$\epsilon_r(\omega) = 1 - \frac{\omega_p^2}{\omega^2}. \quad (7.7)$$

We have assumed here that the light is polarized along the x direction. The model is not affected by this arbitrary choice provided that the medium is isotropic.

We shall see in Section 7.5 that ω_p corresponds to the natural resonant frequency of the whole free carrier gas. This contrasts with the resonant frequency of the individual electrons, which is of course zero because they are free.

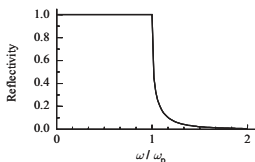


Fig. 7.1 Reflectivity of an undamped free carrier gas as a function of frequency.

The fact that the refractive index is imaginary below ω_p means that the extinction coefficient κ is large, and hence that the medium is highly attenuating. This point will be explained further in Section 7.2. It is a general property of systems with high extinction coefficients that they also have high reflectivities.

The complex refractive index $\tilde{n}$ of the medium is related to the complex dielectric constant by $\tilde{n} = \sqrt{\epsilon_r}$ (cf. eqn 1.22). This means that $\tilde{n}$ is imaginary for $\omega < \omega_p$ and positive for $\omega > \omega_p$, with a value of zero precisely at $\omega = \omega_p$. The reflectivity R can be calculated from eqn 1.29:

$$R = \left| \frac{\tilde{n} - 1}{\tilde{n} + 1} \right|^2. \quad (7.8)$$

By substituting the frequency dependence of $\tilde{n}$ into this formula, we see that R is unity for $\omega \leq \omega_p$, and then decreases for $\omega > \omega_p$, approaching zero at $\omega = \infty$. This frequency dependence is plotted in Fig. 7.1.

The basic conclusion is that we expect the reflectivity of a gas of free electrons to be 100% for frequencies up to ω_p . This result is very well confirmed by experimental data. In Sections 7.3 and 7.4 below we shall see how the plasma reflectivity effect is observed in both metals and doped semiconductors.

One of the best examples of plasma reflectivity effects is the reflection of radio waves from the upper atmosphere. The atoms in the ionosphere are ionized by the ultraviolet light from the Sun to produce a plasma of ions and free electrons. The plasma frequency is in the MHz range, and so the low-frequency waves used for AM radio transmissions are reflected, but not the higher-frequency waves used for FM radio or television. (See Exercise 7.2.)

Example 7.1

Aluminium is a trivalent metal with $6.0 \times 10^{28} \text{ m}^{-3}$ atoms per unit volume. Account for the shiny appearance of aluminium.

Solution

Aluminium has three valence electrons per atom, and so the free electron density N is $3 \times (6.0 \times 10^{28}) = 1.8 \times 10^{29} \text{ m}^{-3}$. We use this value of N in eqn 7.6 to find that $\omega_p = 2.4 \times 10^{16} \text{ rad/s}$. The free electrons in aluminium will reflect all frequencies below ω_p . Now ω_p corresponds to a wavelength of $2\pi c/\omega_p = 79 \text{ nm}$, which is in the ultraviolet spectral region, and this means that all visible wavelengths are reflected. Aluminium therefore has a shiny surface that can be used for making mirrors.

7.2 Free carrier conductivity

In deriving eqn 7.7, we neglected the damping of the free carrier oscillations. We can recast the equation of motion in a way that makes the physical significance of the damping term more apparent. To do this we note that $\dot{\mathbf{x}}$ is the electron velocity $\mathbf{v}$. Hence we can rewrite eqn 7.1 as:

$$m_0 \frac{d\mathbf{v}}{dt} + m_0 \gamma \mathbf{v} = -e\mathcal{E}. \quad (7.9)$$

Since $m_0 \mathbf{v}$ is the momentum $\mathbf{p}$, we see that:

$$\frac{d\mathbf{p}}{dt} = -\frac{\mathbf{p}}{\tau} - e\mathcal{E}, \quad (7.10)$$

where we have replaced the damping rate γ by $1/\tau$, where τ is the damping time. This shows that the electron is being accelerated by the field, but loses its momentum in time τ . In other words, τ is the **momentum scattering time**.

In an AC field of the form $\mathcal{E}(t) = \mathcal{E}_0 e^{-i\omega t}$, we look for solutions to the equation of motion with $\mathbf{x} = \mathbf{x}_0 e^{-i\omega t}$. This implies that $|\mathbf{v}| = \dot{\mathbf{x}}$ also has a time variation of the form $\mathbf{v} = \mathbf{v}_0 e^{-i\omega t}$. On substituting this into eqn 7.9, we obtain:

$$\mathbf{v}(t) = \frac{-e\tau}{m_0} \frac{1}{1 - i\omega\tau} \mathcal{E}(t). \quad (7.11)$$

The current density $\mathbf{j}$ is related to the velocity and field through:

$$\mathbf{j} = -N e \mathbf{v} = \sigma \mathcal{E}, \quad (7.12)$$

where σ is the electrical conductivity. On combining eqns 7.11 and 7.12, we obtain the **AC conductivity** $\sigma(\omega)$:

$$\sigma(\omega) = \frac{\sigma_0}{1 - i\omega\tau}, \quad (7.13)$$

where

$$\sigma_0 = \frac{N e^2 \tau}{m_0}. \quad (7.14)$$

σ_0 is the conductivity measured with DC electric fields. We can thus deduce the momentum scattering time from the DC conductivity through eqn 7.14. For a typical metal or doped semiconductor, this gives values of τ in the range 10^{-14} to 10^{-13} s at room temperature.

By comparing eqns 7.4 and 7.13, we see that the AC conductivity and the dielectric constant are related to each other through:

$$\epsilon_r(\omega) = 1 + \frac{i\sigma(\omega)}{\epsilon_0 \omega}. \quad (7.15)$$

Thus optical measurements of $\epsilon_r(\omega)$ are equivalent to AC conductivity measurements of $\sigma(\omega)$, and the free carrier reflectivity spectrum can be discussed in terms of the conductivity rather than the dielectric constant.

At very low frequencies that satisfy $\omega \ll \tau^{-1}$, we can derive a useful relationship between the conductivity of the free carrier gas and the absorption coefficient for electromagnetic waves. This can be achieved by first splitting $\epsilon_r(\omega)$ into its real and imaginary components in accordance with eqn 1.21, i.e. $\epsilon_r \equiv \epsilon_1 + i\epsilon_2$. Equation 7.5 with $\gamma = \tau^{-1}$ gives:

$$\epsilon_1 = 1 - \frac{\omega_p^2 \tau^2}{1 + \omega^2 \tau^2} \quad (7.16)$$

$$\epsilon_2 = \frac{\omega_p^2 \tau}{\omega(1 + \omega^2 \tau^2)}. \quad (7.17)$$

We then work out n and κ , the real and imaginary parts of the complex refractive index, by using eqns 1.25 and 1.26, and hence deduce the absorption coefficient α from κ . Since $\omega\tau \ll 1$ implies that $\epsilon_2 \gg \epsilon_1$, we can obtain solutions with $n \approx \kappa = (\epsilon_2/2)^{1/2}$. Using eqn 1.19 we therefore obtain:

$$\alpha = \frac{2\omega(\epsilon_2/2)^{1/2}}{c} = \left(\frac{2\omega_p^2 \tau \omega}{c^2} \right)^{1/2}. \quad (7.18)$$

We can put this equation in a more accessible form by noting from eqn 7.14 that $\omega_p^2 \tau = \sigma_0/\epsilon_0$ and from eqn A.28 that $c^2 = 1/\epsilon_0\mu_0$. This gives:

$$\alpha = (2\sigma_0\omega\mu_0)^{1/2}. \quad (7.19)$$

Hence we see that the absorption coefficient is proportional to the square root of the DC conductivity and the frequency.

Equation 7.19 implies that AC electric fields can only penetrate a short distance into a conductor such as a metal. This well-known phenomenon is called the **skin effect**. If the field strength varies as $\exp(-z/\delta)$ with the distance z from the surface, then the power falls off as $\exp(-2z/\delta)$. By comparison with the definition of α in eqn 1.4, we see that:

$$\delta = \frac{2}{\alpha} = \left(\frac{2}{\sigma_0\omega\mu_0} \right)^{1/2}. \quad (7.20)$$

δ is known as the **skin depth**.

The fields that decay exponentially in the conductor are called **evanescent waves**. We have seen in the previous section that we expect the reflectivity of a metal to be very high for frequencies below ω_p . From what we have seen here, it is now apparent that this only applies if the thickness l of the medium is much larger than the skin depth. If l is comparable to, or smaller than, δ , the evanescent fields will not have decayed fully by the back of the medium, and some of the energy will be transmitted. Conservation of energy then demands that the reflectivity R must drop accordingly. This implies that R depends on l when $l \lesssim \delta$, and ultimately drops to zero for a very thin medium.

At higher frequencies the relationship given in eqn 7.19 breaks down because the approximation $\omega\tau \ll 1$ is no longer valid. In this case we can derive a different frequency dependence for the attenuation coefficient. This will be discussed when considering the absorption due to free carriers in doped semiconductors in Section 7.4.1.

A treatment of the variation of R with l may be found, for example, in Born and Wolf (1999).

Example 7.2

The DC electrical conductivity of copper is $6.5 \times 10^7 \Omega^{-1}\text{m}^{-1}$ at room temperature. Calculate the skin depth at 50 Hz and 100 MHz.

Solution

The skin depth is given by eqn 7.20. At a frequency of 50 Hz we have $\omega = 2\pi \times 50 = 314 \text{ rad/s}$. On inserting this value of ω into eqn 7.20 and using $\sigma_0 = 6.5 \times 10^7 \Omega^{-1}\text{m}^{-1}$, we obtain $\delta = 8.8 \text{ mm}$. At 100 MHz, $\omega = 6.28 \times 10^8 \text{ rad/s}$, and the skin depth δ is only $6.2 \mu\text{m}$.

7.3 Metals

The free electron model of metals was proposed by P. Drude in 1900. The model provides a basic explanation for why metals are good conductors of heat and electricity, and is the starting point for more sophisticated theories. As we shall see here, it is also successful in explaining why metals tend to be good reflectors. On the other hand, band theory is needed to explain why some metals (e.g. copper and gold) are coloured.

7.3.1 The Drude model

The Drude free electron model of metals considers the valence electrons of the atoms to be free. When an electric field is applied, the free electrons accelerate and then undergo collisions with the characteristic scattering time τ introduced in eqn 7.10. The electrical conductivity is therefore limited by the scattering, and measurements of σ allow the value of τ to be determined through eqn 7.14.

The free electron density N in the Drude model is equal to the density of metal atoms multiplied by their valency. Table 7.1 lists the Drude free electron densities for a number of common metals. The values of N are in the range 10^{28} – 10^{29} m^{-3} . These very large free electron densities explain why metals have high electrical and thermal conductivities. The plasma frequencies calculated using eqn 7.6 are also tabulated in Table 7.1, together with the wavelength λ_p that corresponds to ω_p . It is apparent that the very large values of N lead to plasma frequencies in the ultraviolet spectral region.

In the visible spectral region where $\omega/2\pi \sim 10^{15} \text{ Hz}$, we are usually in a situation with $\omega \gg \gamma$. This is because $\tau = \gamma^{-1}$ is typically of order 10^{-14} s . Therefore the simplification of eqn 7.5 to eqn 7.7 is a good approximation. With ω_p in the ultraviolet, the visible photons have frequencies below ω_p and thus ϵ_r is negative. As discussed in Section 7.1, this means that the reflectivity is expected to be 100% up to ω_p . This explains the first and most obvious optical property of metals, namely that they tend to be good reflectors at visible frequencies.

Table 7.1 Free electron density and plasma properties of some metals. The figures are for room temperature unless stated otherwise. The electron densities are based on data taken from Wyckoff (1963). The plasma frequency ω_p is calculated from eqn 7.6, and λ_p is the wavelength corresponding to this frequency.

Metal	Valency	N (10^{28} m^{-3})	$\omega_p/2\pi$ (10^{15} Hz)	λ_p (nm)
Li (77 K)	1	4.70	1.95	154
Na (5 K)	1	2.65	1.46	205
K (5 K)	1	1.40	1.06	282
Rb (5 K)	1	1.15	0.96	312
Cs (5 K)	1	0.91	0.86	350
Cu	1	8.47	2.61	115
Ag	1	5.86	2.17	138
Au	1	5.90	2.18	138
Be	2	24.7	4.46	67
Mg	2	8.61	2.63	114
Ca	2	4.61	1.93	156
Al	3	18.1	3.82	79

A striking implication of the free carrier model is that the dielectric constant changes from being negative to positive as we go through the plasma frequency. This means that the reflectivity ceases to be 100% above ω_p (see Fig. 7.1) and some of the light can be transmitted through the metal. Thus we expect that all metals will eventually become transmitting if we go far enough into the ultraviolet so that $\omega > \omega_p$. This phenomenon is known as the **ultraviolet transparency of metals**.

In order to observe the ultraviolet transmission threshold at the plasma frequency, it is necessary that there should be no other absorption processes occurring at ω_p . This condition is best satisfied in the alkali metals. Table 7.2 lists the wavelengths of the ultraviolet transmission edges observed in the alkalis. The experimental wavelengths can be compared with those predicted from the calculated plasma frequency tabulated in Table 7.1. The experimental results are in reasonable agreement with the predictions, and show the correct trend on descending the periodic table. The discrepancies can be explained to a large extent by replacing the free electron mass with the electron effective mass derived from the band structure of the metal. (See Exercise 7.4.)

Figure 7.2 shows the measured reflectivity of aluminium as a function of photon energy from the infrared to the ultraviolet spectral region. As noted in Example 7.1, the plasma frequency occurs in the ultraviolet spectral region, and hence the reflectivity is expected to be high for all visible frequencies. The data show that the reflectivity is over 80% for all photon energies up to $\sim 15 \text{ eV}$, and then drops off to zero at higher energies. Thus aluminium shows the characteristic ultraviolet transparency edge predicted by the Drude model. The relatively featureless reflectivity at visible frequencies is exploited in commercial mirrors.

Table 7.2 Ultraviolet transmission threshold wavelength λ_{UV} for the alkali metals. Data from Givens (1958).

Metal	λ_{UV} (nm)
Li	205
Na	210
K	315
Rb	360
Cs	440

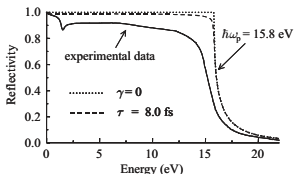


Fig. 7.2 Experimental reflectivity of aluminium as a function of photon energy. The experimental data are compared to predictions of the free electron model with $\hbar\omega_p = 15.8$ eV. The dotted curve is calculated with no damping. The dashed line is calculated with $\tau = 8.0 \times 10^{-15}$ s, which is the value deduced from DC conductivity. Experimental data from Ehrenreich et al. (1963). © American Physical Society, reprinted with permission.

The plasma frequency for aluminium listed in Table 7.1 corresponds to a photon energy of 15.8 eV. The dotted line in Fig. 7.2 gives the reflectivity predicted from eqn 7.7 with $\hbar\omega_p = 15.8$ eV. On comparing the experimental and theoretical results, we see that the model accounts for the general shape of the spectrum, but there are some important details that are not explained.

An improved attempt to model the experimental data can be made by including the damping term in the dielectric constant. Example 7.3 explains how this is done. The reflectivity calculated from eqn 7.5 for the value of τ deduced from the DC conductivity, namely 8.0×10^{-15} s, is plotted as the dashed line in Fig. 7.2. The main difference between the two calculated curves is that the damping causes the reflectivity to be less than unity below ω_p , and the ultraviolet transmission edge is slightly broadened. However, this is only a relatively small effect because $\omega_p \gg \tau^{-1}$.

The inclusion of damping makes a small improvement in the fit to the data, but there are two important features that are still not explained. Firstly, the reflectivity is significantly lower than predicted, and secondly, there is a dip around 1.5 eV, where we would have expected a featureless curve. Both of these points can be explained by considering the interband absorption rates. These are discussed in the next section.

Example 7.3

The conductivity of aluminium at room temperature is $4.1 \times 10^7 \Omega^{-1}\text{m}^{-1}$. Calculate the reflectivity at 500 nm according to the Drude-Lorentz model.

Solution

We first work out the damping time τ from the conductivity using eqn 7.14. Taking the value of $N = 1.81 \times 10^{29} \text{m}^{-3}$ from Table 7.1, we find:

$$\tau = \frac{m_0 \sigma_0}{N e^2} = 8.0 \times 10^{-15} \text{ s}.$$

We shall see in Section 7.5.1 that the plasma frequency can be determined directly by using electron energy-loss spectroscopy.

Table 7.1 also gives us the value of the plasma frequency, namely $\omega_p = 2.4 \times 10^{16}$ rad/s. The wavelength of 500 nm corresponds to an angular frequency $\omega = 2\pi c/\lambda = 3.8 \times 10^{15}$ rad/s. We use these frequencies in eqns 7.16 and 7.17 to calculate the real and imaginary parts of the complex dielectric constant:

$$\epsilon_1 = 1 - \frac{\omega_p^2 \tau^2}{1 + \omega^2 \tau^2} = -39,$$

and

$$\epsilon_2 = \frac{\omega_p^2 \tau}{\omega(1 + \omega^2 \tau^2)} = 1.3.$$

We then work out the real and imaginary parts of the complex refractive index using eqns 1.25 and 1.26. This gives:

$$n = \frac{1}{\sqrt{2}} \left(-39 + [(-39)^2 + (1.3)^2]^{1/2} \right)^{1/2} = 0.10,$$

and

$$\kappa = \frac{1}{\sqrt{2}} \left(+39 + [(-39)^2 + (1.3)^2]^{1/2} \right)^{1/2} = 6.2.$$

We finally obtain the reflectivity from eqn 1.29:

$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} = \frac{(-0.9)^2 + (6.2)^2}{(1.1)^2 + (6.2)^2} = 99\%.$$

This shows that the inclusion of the damping reduces the reflectivity by only 1% in this case.

7.3.2 Interband transitions in metals

The absorption of light by direct interband transitions was discussed in detail in Chapter 3. Direct transitions involve the promotion of electrons to a higher band by absorption of a photon with the correct energy. The electron does not change its k vector significantly because of the very small momentum of the photon. Thus the transitions appear as vertical arrows on the E - k band diagram of the solid.

Interband absorption is important in metals because the electromagnetic waves penetrate a short distance into the surface, and if there is a significant probability for interband absorption, the reflectivity will be reduced. We consider below the reflectivity spectra of aluminium and copper in order to illustrate the effects of interband absorption, and then make some general comments about other metals such as silver and gold.

Aluminium

The band diagram of aluminium is shown in Fig. 7.3. Aluminium has an electronic configuration of $[\text{Ne}]3s^23p^1$ with three valence electrons. The crystal structure is face-centred cubic, which has a body-centred cubic

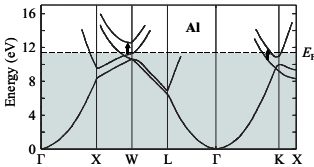


Fig. 7.3 Band diagram of aluminium. The transitions at the W and K points that are responsible for the reflectivity dip at 1.5 eV are labelled. After Segall (1961). © American Physical Society, reprinted with permission.

(bcc) reciprocal lattice, as shown in Fig. D.5 of Appendix D. The first Brillouin zone is completely full, and the valence electrons spread into the second, third and slightly into the fourth zones. The band structure appears quite complex due to the irregular shape of the bcc Brillouin zone. However, the bands are actually very close to the free electron model, with significant departures only in the vicinity of the Brillouin zone boundaries. The bands are filled up to the Fermi energy E_F , which is marked on the diagram. Direct transitions can take place from any of the states below the Fermi level to unoccupied bands directly above them on the E - k diagram.

Fermi's golden rule given in eqn 3.2 tells us that the absorption rate is proportional to the density of states for the transition. The dip in the reflectivity at 1.5 eV which is apparent in Fig. 7.2 is a consequence of the 'parallel band' effect. This occurs when there is a band above the Fermi level that is approximately parallel to another band below E_F . In this case, the interband transitions from a large number of occupied k states below the Fermi level will all occur at the same energy. Hence the density of states at the energy difference between the two parallel bands will be very high, which will result in a particularly strong absorption at this photon energy.

Inspection of the band diagram of aluminium shows that the parallel band effect occurs at both the W and K points of the Brillouin zone. These transitions have been identified on Fig. 7.3. The energy separation of the parallel bands is approximately 1.5 eV in both cases. The enhanced transition rate at this photon energy thus explains the reflectivity dip observed at 1.5 eV in the experimental data. Moreover, we can see from the band diagram that there will be further transitions between bands below the Fermi level to unoccupied bands above E_F at a whole range of photon energies greater than 1.5 eV. The density of states for these transitions will be lower than at 1.5 eV because the bands are not parallel. However, the absorption rate is still significant, and accounts for the reduction of the reflectivity to a value below that predicted by the Drude model in the visible and ultraviolet spectral regions.

We came across a similar example of parallel bands when we discussed the absorption rate at the critical points in the band structure of silicon in Section 3.5.

The positions of the W and K points on the fcc Brillouin zone boundary are shown in Fig. D.5.

Copper

Copper has an electronic configuration of $[\text{Ar}]3d^{10}4s^1$. The outer 4s bands approximate reasonably well to free electron states with a dispersion given by $E = \hbar^2 k^2 / 2m_0$. They therefore form a broad band covering a wide range of energies. The 3d bands, on the other hand, are more tightly bound and are relatively dispersionless, occupying only a narrow range of energies. The density of states for the two bands is illustrated schematically in Fig. 7.4. The narrow 3d bands can hold ten electrons, and therefore their density of states is sharply peaked. The 4s band, which can hold two electrons, is much broader, with a smaller maximum. The 11 valence electrons of copper fill up the 3d band, and half fill the 4s band. The Fermi energy therefore lies within the 4s band above the 3d band. Interband transitions are possible from the filled 3d bands to unoccupied states in the 4s band above E_F , as illustrated in Fig. 7.4. This implies that there will be a well-defined threshold for interband transitions from the 3d bands to the 4s band.

In atomic physics, transitions between d and s states are forbidden for electric-dipole processes. (See Table B.1 in Appendix B.) The matrix element for the 3d $\rightarrow$ 4s transitions is therefore relatively small, but this is compensated by the very high density of states in the solid, which results in strong absorption.

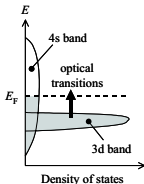


Fig. 7.4 Schematic density of states for the 3d and 4s bands of a transition metal such as copper.

Figure 7.5 shows the actual band structure and density of states of copper. The general features indicated in Fig. 7.4 are clearly shown in the calculated curves. The 4s band is the parabola starting at the Γ point at -9 eV, while the 3d bands are the five curves bunched together in the energy range $-5 \rightarrow -2$ eV. The 4s band crosses the 3d bands and then re-emerges as the single band with energy > -2 eV. It is apparent that the 3d electrons lie in relatively narrow bands with very high densities of states, while the 4s band is much broader with a lower density of states. The Fermi energy lies in the middle of the 4s band above the 3d band. Interband transitions are possible from the 3d bands below E_F to unoccupied levels in the 4s band above E_F . The lowest energy transition is 2.2 eV which corresponds to a wavelength of 560 nm.

Figure 7.6 shows the measured reflectivity of copper from the infrared to the ultraviolet spectral region. Based on the plasma frequency given in Table 7.1, we would expect near 100% reflectivity for photon energies below 10.8 eV, which corresponds to an ultraviolet wavelength of 115 nm. However, the experimental reflectivity falls off sharply above 2 eV owing to the interband absorption edge discussed above. This explains why copper has a reddish colour.

Silver and gold

The arguments used for copper can be applied to the other noble metals. The important parameter is the energy gap between the d bands and the Fermi energy, as shown in Fig. 7.4. In gold the interband absorption threshold occurs at a slightly higher energy than copper, which explains why it has a yellowish colour. In silver, on the other hand, the interband absorption edge is around 4 eV. This frequency is in the ultraviolet, and so the reflectivity remains high throughout the whole visible spectrum. (See Fig. 1.5.) This explains why silver does not have any particular colour, and also why it is so widely used for making mirrors. Gold is also

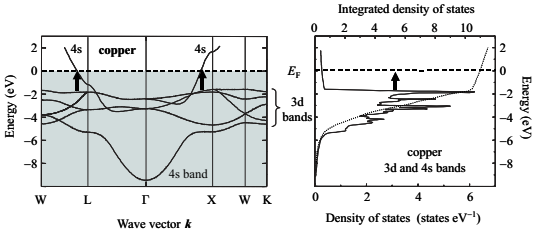


Fig. 7.5 Calculated band structure of copper. The transitions from the 3d bands responsible for the interband transitions around 2 eV are identified. The right-hand side of the figure shows the density of states calculated from the band structure. The strongly peaked features between about -2 eV and -5 eV are due to the 3d bands. The dotted line is the integrated density of states. The Fermi level (defined here as $E = 0$) corresponds to the energy at which the integrated density of states is equal to 11. After Moruzzi et al. (1978).

used for mirrors, but only at infrared wavelengths.

7.4 Doped semiconductors

The controlled doping of semiconductors with impurities is an essential part of solid-state technology. The general principles are discussed in Section D.1 of Appendix D. The introduction of **donor** impurities provides an excess of electrons, while **acceptor** impurities lead to a deficit of electrons, which is equivalent to an excess of holes. Doping that produces excess electrons is called **n-type**, while doping that produces excess holes is called **p-type**.

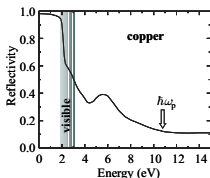
Experimental measurements on doped semiconductors show that the presence of impurities gives rise to new absorption mechanisms and also to a free carrier plasma reflectivity edge. Our aim here is to explain these effects by applying a suitably modified version of the free carrier model and by considering the quantized levels created by the impurity atoms. In the two subsections that follow, we first consider the free carrier effects, and then move on to discuss the absorption associated with the impurity levels.

7.4.1 Free carrier reflectivity and absorption

The free electron model developed in Sections 7.1 and 7.2 can be applied to doped semiconductors if we make two appropriate modifications. Firstly, we must account for the fact that the electrons and holes

In silicon and germanium, which come from group IV of the periodic table, n-type doping is achieved by adding atoms from group V, while p-type doping is achieved by adding atoms from group III. In compound semiconductors such as the III-Vs, the way the doping works is more complicated. If the impurity sits on the group III atom site, then a group II element gives p-type doping, and a group IV element gives n-type. On the other hand, if the impurity sits on the group V atom site, then a group IV impurity gives p-type doping, while a group VI element gives n-type. Group IV doping of a III-V semiconductor can therefore give either n- or p-type doping, depending on how the impurities fit into the crystal.

Fig. 7.6 Reflectivity of copper from the infrared to the ultraviolet spectral region. The reflectivity drops sharply above 2 eV due to interband transitions. Data from Lide (1996).



are moving in the conduction or valence band of a semiconductor. This is easily achieved by assuming that the carriers behave as particles with an effective mass m^* rather than the free electron mass m_0 . Secondly, we must remember that the semiconductor has a high relative permittivity at the frequencies of interest even before the dopants are added.

The two modifications mentioned above can be handled if we rewrite eqn 7.3 in the following form:

$$\begin{aligned} D &= \epsilon_r \epsilon_0 \mathcal{E} \\ &= \epsilon_0 \mathcal{E} + P_{\text{other}} + P_{\text{free carrier}} \\ &= \epsilon_{\text{opt}} \epsilon_0 \mathcal{E} - \frac{Ne^2 \mathcal{E}}{m^* (\omega^2 + i\gamma\omega)}. \end{aligned} \quad (7.21)$$

The term P_{other} accounts for the polarizability of the bound electrons before the dopants are added, while the effective mass m^* accounts for the band structure of the semiconductor. The carrier density N that appears in this equation is the density of free electrons or holes generated by the doping process. Note that the sign of the charge cancels, and so the only difference between electrons and holes in this treatment is in the effective mass that is used.

The free carrier effects due to doping are most noticeable in the spectral region 5–30 μm , where we would normally expect the semiconductor to be completely transparent. Hence the value of ϵ_{opt} that we use in eqn 7.21 is the one measured in the transparent spectral region below the interband absorption edge. This value is known from the refractive index of the undoped semiconductor: $\epsilon_{\text{opt}} = n^2$. (See eqn 1.27 with $\kappa = 0$ below the band edge.)

Equation 7.21 tells us that the frequency dependence of the dielectric constant is given by:

$$\epsilon_r(\omega) = \epsilon_{\text{opt}} - \frac{Ne^2}{m^* \epsilon_0} \frac{1}{(\omega^2 + i\gamma\omega)}. \quad (7.22)$$

This can be rewritten as:

$$\epsilon_r(\omega) = \epsilon_{\text{opt}} \left(1 - \frac{\omega_p^2}{(\omega^2 + i\gamma\omega)} \right), \quad (7.23)$$

As explained in Section 2.2.2 of Chapter 2, solids have a number of resonant frequencies, each of which can be modelled by dipole oscillators. There are resonant frequencies in the infrared due to the phonons, and others in the near-infrared, visible, or ultraviolet due to the bound electrons. The phonon absorption bands are discussed in detail in Chapter 10, and occur in the range 30–100 μm for a typical III–V semiconductor.

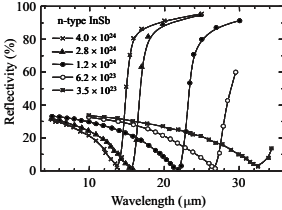


Fig. 7.7 Infrared reflectivity spectra of n-type InSb at room temperature for different values of the free electron density. After Spitzer & Fan (1957). © American Physical Society, reprinted with permission.

where the plasma frequency ω_p is now given by:

$$\omega_p^2 = \frac{Ne^2}{\epsilon_{\text{opt}}\epsilon_0 m^*}. \quad (7.24)$$

We have written the dielectric constant in this way to make the link to the Drude model apparent. The difference between the plasma frequency for the semiconductor given in eqn 7.24 and that given in eqn 7.6 is that we have replaced m_0 by m^* , and we have included ϵ_{opt} to account for the background polarizability of the undoped semiconductor.

If we assume that the system is lightly damped, then we can ignore the damping term in eqn 7.23. This then implies that ϵ_r is negative below ω_p and positive at higher frequencies. We thus expect to observe a plasma reflectivity edge at ω_p just as we did in metals. Since the carrier density is much smaller than in metals, the plasma edge occurs at frequencies in the infrared spectral range. This prediction is very well borne out by infrared reflectivity data.

Figure 7.7 shows the measured reflectivity of n-type InSb as a function of the electron density. The fundamental absorption edge at the band gap of InSb occurs at 6 μm , while the phonon absorption band lies around 50 μm . Thus we would expect pure InSb to be transparent in the wavelength range shown and have a featureless reflectivity spectrum. Instead, the data show a well defined reflectivity edge, which shifts to shorter wavelengths as the electron density increases, in accordance with eqn 7.24.

The data shown in Fig. 7.7 demonstrate the phenomenon of the plasma reflectivity edge more clearly than many of the equivalent results obtained for metals. This is because it is not possible to vary the electron density in metals. Moreover, in metals the plasma frequencies are much higher, and the reflectivity edge is frequently obscured by interband transitions.

One very striking feature of the data is the zero in the reflectivity at wavelengths just below the plasma edge. This occurs at a frequency

given by (see Exercise 7.8):

$$\omega^2 = \frac{\epsilon_{\text{opt}}}{\epsilon_{\text{opt}} - 1} \omega_p^2. \quad (7.25)$$

By fitting this formula to the data, the effective mass of InSb can be determined. (See Exercise 7.9.)

At frequencies above ω_p , the presence of free carriers leads to the absorption of light. This effect is called **free carrier absorption**, and can be observed in the infrared spectral region below the fundamental absorption edge at the band gap, where the semiconductor would normally be transparent. To see how this effect arises, we split the dielectric constant given in eqn 7.23 into its real and imaginary parts. This gives:

$$\epsilon_1 = \epsilon_{\text{opt}} \left(1 - \frac{\omega_p^2 \tau^2}{1 + \omega^2 \tau^2} \right), \quad (7.26)$$

$$\epsilon_2 = \frac{\epsilon_{\text{opt}} \omega_p^2 \tau}{\omega(1 + \omega^2 \tau^2)}, \quad (7.27)$$

where we have made the usual substitution of τ^{-1} for γ . In a typical semiconductor, with $\tau \sim 10^{-13}$ s at room temperature, it is safe to make the approximation $\omega\tau \gg 1$ at frequencies in the near-infrared. Furthermore, the free carrier term in ϵ_r will be small. Therefore we can assume $\epsilon_1 \approx \epsilon_{\text{opt}}$, and that $\epsilon_2 \ll \epsilon_1$. In these conditions we find solutions to eqns 1.25 and 1.26 with $n = \sqrt{\epsilon_{\text{opt}}}$ and $\kappa = \epsilon_2/2n$. This allows us to deduce the absorption coefficient using eqn 1.19. The result is:

$$\alpha_{\text{free carrier}} = \frac{\epsilon_{\text{opt}} \omega_p^2}{nc\omega^2 \tau} = \frac{N e^2}{m^* \epsilon_0 n c \tau} \frac{1}{\omega^2}. \quad (7.28)$$

This shows that the free carrier absorption is proportional to the carrier density and should vary with frequency as ω^{-2} .

Experimental data on a number of n-doped samples lead to the conclusion that $\alpha_{\text{free carrier}} \propto \omega^{-\beta}$, where β is in the range 2–3. The departure of β from the predicted value of 2 is caused by the failure of our assumption that τ is independent of ω . To see why this is important, we illustrate the physical processes that are occurring during free carrier absorption in Fig. 7.8. The figure shows the conduction band of an n-type semiconductor, which is filled up to the Fermi level determined by the free carrier density. Absorption of a photon excites an electron from an occupied state below the Fermi level to an unoccupied level above E_F . The photon only has a very small momentum compared to the electron, and therefore cannot change the electron's momentum significantly. It is obvious from Fig. 7.8 that a scattering event must occur to conserve momentum in the process. Hence the absorption must be proportional to the scattering rate $1/\tau$, in accordance with the prediction of eqn 7.28.

The mechanisms that can contribute to the momentum-conserving process in free carrier absorption include phonon scattering and scattering from the ionized impurities left behind by the release of the free

The skin effect considered in Section 7.2 may also be considered as a type of free carrier absorption. In the skin effect, however, we are considering the absorption at low frequencies below ω_p where the material is highly reflective. We are now considering absorption above ω_p where the material should be transparent.

The approximation that τ is independent of ω effectively says that the relaxation time of the electrons does not depend on their initial energy. This is equivalent to the energy independent relaxation time approximation of the Boltzmann equation used in electron transport theory. It is well known that this approximation is only valid in a limited range of conditions. See Ashcroft and Mermin (1976) for further details.

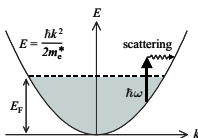


Fig. 7.8 A free carrier transition in a doped semiconductor.

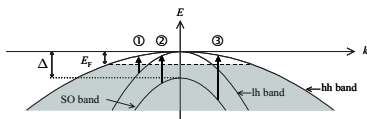


Fig. 7.9 Intervalence band absorption in a p-type semiconductor. E_F is the Fermi energy determined by the doping density. The labelled arrows indicate: (1) transitions from the light-hole (lh) band to the heavy-hole (hh) band; (2) transitions from the split-off (SO) band to the lh band; and (3) transitions from the SO band to the hh band.

electrons from their dopants. It is a sweeping oversimplification to characterize all the possible scattering processes with a single frequency-independent scattering time τ deduced from the DC conductivity. Thus it is hardly surprising that the experimental data do not exactly show an ω^{-2} dependence.

The free carrier reflectivity and absorption of p-type semiconductors can be modelled by a similar treatment to the one developed here for n-type samples. The only change that has to be made is in the effective mass that is used in the calculation. Thus we would expect that all the main results will hold, provided we take account of the fact that the scattering time for holes is not necessarily the same as that for electrons. However, p-type samples also show another effect, which is discussed next.

Figure 7.9 shows the valence band of a p-type III-V semiconductor. This is a larger scale version of the band structure diagram given previously in Fig. 3.5, except that now there are unfilled states near $k = 0$ owing to the p-type doping. Optical transitions can take place in which an electron is promoted from an occupied state below E_F in the light-hole (lh) band to an empty one in the heavy-hole (hh) band above E_F . This is called **intervalence band absorption**. Other intervalence band transitions are possible in which an electron is promoted from the split-off (SO) band to either the lh or hh band. The range of energies over which these transitions occur can be calculated from the effective masses, the doping density and the spin-orbit energy Δ (see Exercise 7.12). The absorption occurs in the infrared, and measurements of the spectrum can give values for Δ and the ratio of the hole effective masses. The absorption can be a strong process because no scattering events are required

Note that intervalence band transitions are forbidden at $k = 0$ since all the hole bands are derived from p-like atomic states. The atomic character of the bands is less well defined for finite k , and this makes the transitions possible away from the centre of the Brillouin zone.

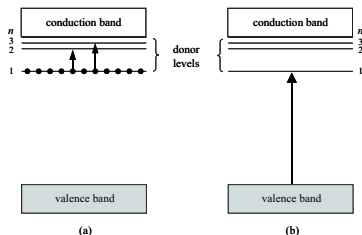


Fig. 7.10 Impurity absorption mechanisms in an n-type semiconductor: (a) transitions between donor levels; (b) transitions from the valence band to empty donor levels. The donor level energy spacings have been exaggerated in this diagram to make the mechanisms clearer.

to conserve momentum.

7.4.2 Impurity absorption

The n-type doping of a semiconductor with donor atoms introduces a series of hydrogenic levels just below the conduction band. These quantized states are called **donor levels**, and are illustrated schematically in Fig. 7.10. The impurity levels give rise to two new absorption mechanisms, in addition to the free carrier effects discussed in the previous section. If the donor states are occupied, it will be possible to absorb photons by exciting electrons between the levels as illustrated in Fig. 7.10(a). On the other hand, if the states are empty, then it will be possible to absorb light by exciting electrons from the valence band to the donor states as illustrated in Fig. 7.10(b).

We consider first the transitions between the donor levels illustrated in Fig. 7.10(a). For such a process to occur, the donor levels must be occupied. This will be the case at low temperatures, when there is insufficient thermal energy to promote the electrons from the donor levels into the conduction band.

The frequencies of the donor-level transitions can be calculated if the energies of the impurity states are known. In the simplest model, we assume that the electron is released into the crystal, and is then attracted back towards the positively charged impurity atom. The electron and the ionized impurity then form a hydrogenic system bound together by their mutual Coulomb attraction. As a first approximation, we can use the Bohr formula, provided we use the effective mass m_e^* instead of the free electron mass m_0 , and also include the dielectric constant ϵ_r for the semiconductor. Hence the energy of the donor levels E_n^D will be given by:

$$E_n^D = -\frac{m_e^*}{m_0} \frac{1}{\epsilon_r^2} \frac{R_H}{n^2}, \quad (7.29)$$

Equation 7.29 is very similar to eqn 4.1 for the exciton binding energy except that the electron effective mass appears instead of the reduced electron-hole mass. This is because we are now considering the attraction of an electron to a heavy ion which is bound in the lattice, instead of that between a free electron and a free hole.

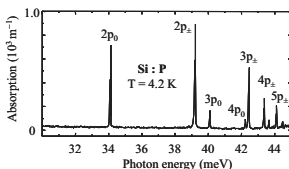


Fig. 7.11 Infrared absorption spectrum of n-type silicon doped with phosphorus at a density of $1.2 \times 10^{20} \text{ m}^{-3}$. The temperature was 4.2 K. After Jagannath et al. (1981), © American Physical Society, reprinted with permission.

where R_H is the hydrogen Rydberg energy (13.6 eV) and n is an integer.

At low temperatures we can assume that all the electrons from the donors will be in the $n = 1$ ground state impurity level. Optical transitions can then take place in which the electrons are promoted to higher donor levels or into the conduction band by absorption of a photon. Figure 7.10(a) illustrates two possible transitions of this type, in which the electron is promoted to either the $n = 2$ or the $n = 3$ donor level. These transitions give rise to absorption lines analogous to the hydrogen Lyman series with frequencies given by:

$$h\nu = \frac{m_e^*}{m_0} \frac{R_H}{\epsilon_f^2} \left(1 - \frac{1}{n^2} \right), \quad (7.30)$$

where n is the quantum number of the final impurity level. If we insert typical values into eqn 7.30 we find that the photon energies are in the range 0.01–0.1 eV. This means that the transitions occur in the infrared spectral region.

Figure 7.11 shows the absorption spectrum of n-type silicon at liquid helium temperatures. The sample was doped with phosphorus at a density of $1.2 \times 10^{20} \text{ m}^{-3}$. The absorption lines correspond to transitions exciting electrons from the $n = 1$ shell to higher shells. In the language of atomic physics, these are $1s \rightarrow np$ transitions. These transitions converge at high n to the donor ionization energy of phosphorus in silicon, which is 45 meV.

The spectrum shown in Fig. 7.11 is actually more complicated than eqn 7.30 would suggest. It consists of two series of transitions, which are labelled as either np_0 or $np_{\pm}$. The np_0 series obey eqn 7.30 very well, but the $np_{\pm}$ transitions have a different frequency dependence. This complexity is caused by the anisotropy of the effective mass of silicon. The frequency dependence of the two series can be modelled by assigning different effective Rydberg energies for the '0' and ' $\pm$ ' states. (See Exercise 7.13.)

We now consider the absorption mechanism shown in Fig. 7.10(b). These transitions can be observed at temperatures when the donor levels are partly unoccupied owing to the thermal excitation of the electrons into the conduction band. Absorption processes can then occur in which

The absolute value of the absorption coefficient for the impurity transitions in Fig. 7.11 is around 10^3 m^{-1} . This is much smaller than for interband transitions which typically have values in the range 10^6 – 10^8 m^{-1} . However, if we were to assume that the absorption strength is simply proportional to the number of atoms that contribute, we would expect the impurity absorption to be weaker than the interband absorption by about a factor of $\sim 10^9$. The measured ratio is much larger because the impurity lines are very sharp, whereas the interband transitions spread out into bands.

In many direct-gap semiconductors it is found experimentally that the absorption decreases exponentially below the band gap. This is called the Urbach tail on account of **Urbach's rule**, which states that the frequency dependence of the absorption for $\hbar\omega < E_g$ is given by:

$$\alpha(\hbar\omega) \propto \exp\left(\frac{\sigma(\hbar\omega - E_g)}{k_B T}\right),$$

where σ is a phenomenological fitting parameter.

electrons are excited from the top of the valence band to the empty donor levels.

The valence band $\rightarrow$ donor level transitions occur at photon energies just below the band gap E_g , with a threshold given by $E_g - E_I^D$. However, the transitions tend to be broadened into a continuum both by thermal effects and by the fact that transitions can take place from a whole range of states within the valence band. Hence the impurity transitions cause a smearing of the absorption edge compared with the abrupt edge found at the band gap of pure semiconductors. The absorption strength will always be weak compared to the interband and excitonic transitions due to the relatively small number of impurity atoms compared to the density of states within the conduction band. On the other hand, the transitions occur in the spectral region just below the band gap where we would normally expect no absorption at all. Hence these transitions do have an effect on the fundamental absorption edge, and make precise determinations of E_g from the absorption spectra at room temperature more difficult.

We have restricted our attention here to n-type semiconductors for the sake of simplicity. The same effects can of course occur in p-type materials.

7.5 Plasmons

Equation 7.7 tells us that the relative permittivity of a lightly damped gas of free electrons is expected to be zero at ω_p . This suggests that something interesting might happen at this frequency. This is indeed the case, as we discuss here.

7.5.1 Bulk plasmons

A plasma consists of a gas of charged particles in dynamic equilibrium. The particles are in constant motion, and this can create local charge fluctuations. If a fluctuation were to create a small region with an excess charge, the charges in that volume would be repelled away by the surrounding charges. The velocity acquired in this process could cause the excess charges to overshoot their original position, in which case they would then be pushed back in the opposite direction. This process can lead to oscillatory motion called **plasma oscillations**. These plasma oscillations are well known in gas discharge tubes, and they can also occur in the free electron plasmas found in metals and doped semiconductors, which is our interest here.

The frequency of the plasma oscillations can be calculated as follows. Consider a region of a conducting medium of volume V enclosed by a surface S . Conservation of charge requires that the net flow of current into or out of a region must be balanced by the change of the total charge

inside the surface. This continuity condition can be written:

$$\oint_S \mathbf{j} \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_V \rho dV, \quad (7.31)$$

where $\mathbf{j}$ is the current density, $d\mathbf{S}$ is a surface element, ρ is the local charge density, and dV is a volume element. On applying the divergence theorem, we find that:

$$\int_V \nabla \cdot \mathbf{j} dV = -\int_V \frac{\partial \rho}{\partial t} dV, \quad (7.32)$$

and hence (since the volume integrated over is arbitrary):

$$\nabla \cdot \mathbf{j} = -\frac{\partial \rho}{\partial t}. \quad (7.33)$$

Equation 7.33 is called the **charge continuity equation**.

We now consider the case in which we have a collective motion of the free electrons relative to the fixed lattice of positive ions in a metal or doped semiconductor. The overall charge density is zero, but the movement of the electrons can create local currents. Since the positive charges on the ions are stationary, they do not generate a current, and we can apply the charge continuity equation to the electron current alone, giving

$$\nabla \cdot \mathbf{j} = -\frac{\partial \rho_e}{\partial t}, \quad (7.34)$$

where ρ_e is the electron charge density. Then, on substituting for ρ_e from Gauss's law (i.e. $\nabla \cdot \mathbf{E} = \rho_e/\epsilon_0$), we find:

$$\nabla \cdot \left(\mathbf{j} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = 0. \quad (7.35)$$

Now a vector that has zero divergence can always be written as the curl of another vector, and from Maxwell's fourth equation (eqn A.13), we realize that this vector must be $\mathbf{B}/\mu_0$, giving:

$$\mathbf{j} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\mu_0} \nabla \times \mathbf{B}. \quad (7.36)$$

On taking the time derivative and substituting from Maxwell's third equation (eqn A.12), we find:

$$\begin{aligned} \frac{\partial \mathbf{j}}{\partial t} + \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} &= \frac{1}{\mu_0} \nabla \times \frac{\partial \mathbf{B}}{\partial t} \\ &= -\frac{1}{\mu_0} \nabla \times (\nabla \times \mathbf{E}). \end{aligned} \quad (7.37)$$

The electrons will move in response to the local electric field according to their equation of motion:

$$m\dot{\mathbf{v}} = -e\mathbf{E}. \quad (7.38)$$

The divergence theorem of mathematics requires that:

$$\oint_S \mathbf{j} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{j} dV,$$

where the volume integral is over the region enclosed by the surface S .

Now the current density is given by $\mathbf{j} = -Ne\mathbf{v}$, which implies that

$$\frac{\partial \mathbf{j}}{\partial t} = \frac{Ne^2}{m} \mathbf{E}. \quad (7.39)$$

On substituting into eqn 7.37 and re-arranging, we find:

$$\frac{\partial^2 \mathbf{E}}{\partial t^2} + \omega_p^2 \mathbf{E} = -c^2 \nabla \times (\nabla \times \mathbf{E}), \quad (7.40)$$

where we have substituted for ω_p from eqn 7.6 and used $c^2 = 1/\mu_0\epsilon_0$ (cf. eqn A.28).

At this stage it is helpful to split the electric field into transverse and longitudinal components:

$$\mathbf{E} = \mathbf{E}_t + \mathbf{E}_l, \quad (7.41)$$

where $\nabla \cdot \mathbf{E}_t = 0$ and $\nabla \times \mathbf{E}_l = 0$. On substituting into eqn 7.40 and using the vector identity of eqn A.24, this gives:

$$\frac{\partial^2 \mathbf{E}_t}{\partial t^2} + \omega_p^2 \mathbf{E}_t - c^2 \nabla^2 \mathbf{E}_t = - \left(\frac{\partial^2 \mathbf{E}_l}{\partial t^2} + \omega_p^2 \mathbf{E}_l \right). \quad (7.42)$$

It becomes apparent that both sides of eqn 7.42 must be equal to zero by taking the divergence and curl of the equation.

This implies that we have two independent equations of motion for the transverse and longitudinal components:

$$\frac{\partial^2 \mathbf{E}_t}{\partial t^2} + \omega_p^2 \mathbf{E}_t - c^2 \nabla^2 \mathbf{E}_t = 0, \quad (7.43)$$

$$\frac{\partial^2 \mathbf{E}_l}{\partial t^2} + \omega_p^2 \mathbf{E}_l = 0. \quad (7.44)$$

We look for wave-like solutions in which the field varies with time and position as $\exp(i(\mathbf{k} \cdot \mathbf{r} - \omega t))$. For the transverse solutions we find:

$$c^2 k^2 = \omega^2 - \omega_p^2. \quad (7.45)$$

This describes the dispersion of conventional transverse electromagnetic waves in the plasma. The dispersion is plotted in Fig. 7.16(a) below. There are no travelling solutions with $\omega < \omega_p$ because the waves are reflected by the plasma.

In the case of the longitudinal modes, we simply have:

$$\omega = \omega_p. \quad (7.46)$$

In a metal, the frequency of the longitudinal modes does in fact vary slightly with k . The correction term is very small, and arises from the breakdown of some of the approximations used in this derivation. See Exercise 7.18.

This shows that the medium can support longitudinal modes at the plasma frequency, and that the modes are dispersionless: i.e. ω is independent of k . These longitudinal modes correspond to the plasma oscillations that we discussed qualitatively at the start of the section. Figure 7.12(a) shows a schematic diagram of the longitudinal electron displacements within a plasma oscillation and the fields that they generate.

The existence of longitudinal solutions at the plasma frequency is a consequence of the fact that $\epsilon_r = 0$ at ω_p . In a medium with zero average

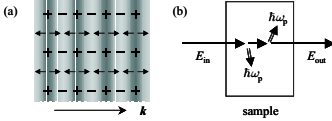


Fig. 7.12 (a) Charge fluctuations in a free carrier plasma oscillation. The lighter regions denote areas with excess electron densities. The small arrows indicate the direction of the electric fields, which are parallel to the direction of propagation of the wave, as indicated by its wave vector $\mathbf{k}$. (b) Excitation of plasmons by inelastic scattering of particles. The case in which two plasmons are excited is shown. For metals, electrons with keV energies are used, but for doped semiconductors, optical frequency photons have sufficient energy.

charge density, Gauss's law (eqn A.10) combined with eqn A.3 tells us that

$$\nabla \cdot \mathbf{D} = \nabla \cdot (\epsilon_r \epsilon_0 \mathbf{E}) = 0. \quad (7.47)$$

If $\epsilon_r \neq 0$, we then deduce that $\nabla \cdot \mathbf{E} = 0$. This is the normal situation for transverse electromagnetic waves in which the electric field is perpendicular to the direction of the wave. However, if $\epsilon_r = 0$, we can satisfy eqn 7.47 with waves that have $\nabla \cdot \mathbf{E} \neq 0$, i.e. longitudinal waves. We thus conclude that a dielectric can support longitudinal electric field waves at frequencies that satisfy $\epsilon_r(\omega) = 0$.

Equation 7.44 shows us that the longitudinal oscillations of the plasma behave as harmonic oscillators with a natural resonant frequency at ω_p . The derivation is completely classical, and the oscillator can have any energy. However, we know in fact that the energy of harmonic oscillators is quantized. We therefore expect the energy of the plasma oscillations to be quantized in units of $\hbar\omega_p$. The quasi-particles that correspond to these quantized plasma oscillations are called **plasmons**. As shown in eqn 7.46, the frequency of the plasmons is independent of their wave vector.

Since plasmons are associated with longitudinal plasma oscillations, they cannot be excited directly by light, which is a transverse wave. Instead, they have to be observed by techniques of inelastic scattering, in which a beam of particles excites plasmons while passing through the medium, as illustrated in Fig. 7.12(b). The energy E_{in} of the incoming particles must be significantly larger than the plasmon energy. Conservation of energy requires that:

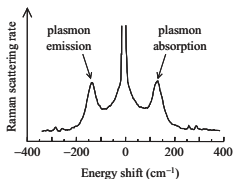
$$E_{\text{out}} = E_{\text{in}} - n\hbar\omega_p, \quad (7.48)$$

where E_{out} is the energy of the outgoing particle, and n is the number of plasmons emitted. The detection of particles with energies given by eqn 7.48 establishes that plasmons have been excited.

In the case of metals, the plasmon energies are several eV, and so electrons with keV energies are typically used. By measuring the energy spectrum of the electrons emerging from a thin sample, the plasma

We shall come across another example of longitudinal modes at frequencies where $\epsilon_r = 0$ when we consider phonons in Chapter 10. (See Section 10.2.2.)

Fig. 7.13 Raman scattering measurements on n-type GaAs at 300 K. The doping density was $1.75 \times 10^{23} \text{ m}^{-3}$. The data are displayed as a function of the energy shift of the outgoing photons relative to the incoming ones in wave number units. After Mooradian (1972), © Excerpta Medica Inc., reprinted with permission.



frequency can be determined. This technique is called **electron energy-loss spectroscopy**.

Plasmons can also be observed in doped semiconductors. Since the plasma frequencies are much lower, it is possible to use inelastic light scattering techniques (i.e. Raman scattering) to measure the plasmon energies. The general principles of Raman scattering will be discussed in Section 10.5. The basic point is that the energy $\hbar\omega_{\text{out}}$ of the outgoing photon must satisfy:

$$\hbar\omega_{\text{out}} = \hbar\omega_{\text{in}} \pm \hbar\omega_p, \quad (7.49)$$

where $\hbar\omega_{\text{in}}$ is the energy of the incoming photon. The + sign corresponds to plasmon absorption and the - sign to plasmon emission. Plasmon absorption is possible here, but not in the case of metals, because the plasmon energies are comparable to the thermal energy $k_B T$. This means that there might already be plasmons excited in the sample before the incident photon arrives, and thus there is some probability that a plasmon might be destroyed and its energy transferred to the photon.

Figure 7.13 shows the results of a Raman scattering experiment on n-type GaAs at 300 K. The doping density was $1.75 \times 10^{23} \text{ m}^{-3}$. The Raman intensity is plotted as a function of the frequency shift of the light in wave number units. The data show two clear peaks shifted by $\pm 130 \text{ cm}^{-1}$ relative to the incoming laser beam due to plasmon emission and absorption. The electron effective mass of GaAs is $0.067m_0$ and ϵ_{opt} is 10.6. Hence from eqn 7.24 we find $\omega_p = 2.8 \times 10^{13} \text{ rad/s}$, which is equivalent to 150 cm^{-1} . The experimental data are thus in reasonably good agreement with the model.

7.5.2 Surface plasmons

Careful analysis of the electron energy-loss spectra from a metal typically reveals that there are two different types of plasmon within the metal, namely bulk and surface plasmons. We have considered the first type in the previous subsection, and our task now is to explain the second. Interest in these surface plasmons has increased dramatically in recent years, and a new field of research has burgeoned called **plasmonics**.

Two weak peaks at $\pm 272 \text{ cm}^{-1}$ and $\pm 296 \text{ cm}^{-1}$ are also present in Fig. 7.13. These are caused by optical phonons. (See Section 10.5.2.) The phonon signals are linearly polarized, and have been strongly suppressed in the data by the use of orthogonal polarizers in front of the detector. Note that it is very common to use wave number units in Raman spectroscopy. The wave number $\bar{\nu}$ is equal to the reciprocal of the wavelength: $\bar{\nu} = 1/\lambda$. It is effectively a unit of energy with $1 \text{ cm}^{-1} \equiv 0.124 \text{ meV}$.

Surface plasmons are quantized electromagnetic surface waves that are localized at the interface between a plasma and a dielectric material. We are interested here in the case where the plasma is a metal. The dielectric is usually the air, although it might also typically be glass or a semiconductor. The waves propagate along the interface plane, and the electron charge density fluctuations in the metal generate electric field lines as shown in Fig. 7.14. The amplitude of the electric field decays exponentially on either side of the interface.

It is apparent from Fig. 7.14 that the surface plasmons have both transverse and longitudinal electric field components, which contrasts with bulk plasmons which are purely longitudinal. The presence of the transverse component means that surface plasmons can interact directly with photons. This interaction is sufficiently strong that we need to consider the photon and plasmon as a coupled system called a **polariton**. In general, polaritons are coupled electric *polarization*–photon waves. Several different types of polariton are possible, and the type of polariton that we are considering here is called a **surface plasmon polariton**.

The dispersion of the surface plasmon polaritons can be found by solving Maxwell's equations. We define axes so that the plane $z = 0$ corresponds to the interface, with positive and negative z corresponding to the dielectric and metal respectively, as shown in Fig. 7.15(a). The wave is assumed to be propagating in the x direction. The electric field has components in both the x and z directions, and its amplitude decays exponentially as a function of the distance from the interface, as shown in Fig. 7.15(b). In this geometry, the electric and magnetic fields can be written in the form:

$$\begin{aligned}\mathcal{E}^d(x, z, t) &= [\mathcal{E}_x^d, 0, \mathcal{E}_z^d] e^{i(k_x^d x - \omega t)} e^{-k_z^d z}, \\ \mathbf{H}^d(x, z, t) &= [0, H_y^d, 0] e^{i(k_x^d x - \omega t)} e^{-k_z^d z}, \\ \mathcal{E}^m(x, z, t) &= [\mathcal{E}_x^m, 0, \mathcal{E}_z^m] e^{i(k_x^m x - \omega t)} e^{+k_z^m z}, \\ \mathbf{H}^m(x, z, t) &= [0, H_y^m, 0] e^{i(k_x^m x - \omega t)} e^{+k_z^m z},\end{aligned}\quad (7.50)$$

where the labels 'd' and 'm' refer to the dielectric and metal respectively. Note that the opposite sign of the decay term in the z direction in the dielectric and the metal ensures that the fields are localized at the interface.

The fields given in eqn 7.50 must satisfy Maxwell's equations and the boundary conditions that apply when there is no net free charge density. Consider first the boundary conditions. The tangential components of $\mathcal{E}$ and $\mathbf{H}$, together with the normal component of the electric displacement $\mathbf{D}$, must be continuous at the interface. On recalling that $\mathbf{D} = \epsilon_r \epsilon_0 \mathcal{E}$, we then have that:

$$\begin{aligned}\mathcal{E}_x^d &= \mathcal{E}_x^m, \\ H_y^d &= H_y^m, \\ \epsilon_d \mathcal{E}_z^d &= \epsilon_m \mathcal{E}_z^m,\end{aligned}\quad (7.51)$$

where ϵ_d and ϵ_m are the relative permittivities of the dielectric and metal respectively. The requirement that these conditions apply along all the

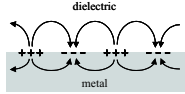


Fig. 7.14 Electric fields associated with electron charge density fluctuations at the surface of a metal.

Phonon polaritons are considered in Section 10.3.

In principle, it might also be possible to have waves with electric and magnetic field components along the y and x directions respectively. However, it can be shown that these solutions are not possible. See Maier (2007).

Note that the most interesting case to consider is when $\omega < \omega_p$, where ϵ_m is *negative*. $\mathcal{E}_x^d$ and $\mathcal{E}_x^m$ therefore point in opposite directions, as shown in Fig. 7.14.

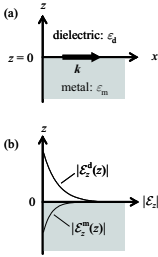


Fig. 7.15 (a) Definition of axes for the interface between a dielectric medium and a metal with relative permittivities of ϵ_d and ϵ_m respectively. The plane $z = 0$ defines the interface, and the polariton propagates in the x direction, as shown by the $\mathbf{k}$ vector. (b) Exponential decay of the field amplitudes as a function of the distance from the interface.

surface implies that

$$k_x^d = k_x^m \equiv k_x, \quad (7.52)$$

where k_x is the common x component of the wave vector on both sides of the interface.

Now consider Maxwell's fourth equation (eqn A.13) with $\mathbf{j} = 0$ and $\mathbf{D} = \epsilon_r \epsilon_0 \mathcal{E}$. The fact that $\mathcal{E}_y = 0$ and $H_x = 0$ allows us to relate H_y to $\mathcal{E}_x$:

$$-\frac{\partial H_y}{\partial z} = \epsilon_r \epsilon_0 \frac{\partial \mathcal{E}_x}{\partial t}. \quad (7.53)$$

On substituting the fields from eqn 7.50, this gives:

$$\begin{aligned} k_z^d H_y^d &= -i \epsilon_d \epsilon_0 \omega \mathcal{E}_x^d, \\ -k_z^m H_y^m &= -i \epsilon_m \epsilon_0 \omega \mathcal{E}_x^m. \end{aligned} \quad (7.54)$$

By using $\mathcal{E}_x^d = \mathcal{E}_x^m$ and $H_y^d = H_y^m$ from eqn 7.51, we can rearrange this to find:

$$\frac{k_z^d}{\epsilon_d} + \frac{k_z^m}{\epsilon_m} = 0. \quad (7.55)$$

Note that this is consistent with both k_z^d and k_z^m being positive when ϵ_m is negative, i.e. when $\omega < \omega_p$. Since the overall charge density is zero, the fields must satisfy (cf. eqns A.25 and A.28 with $\mu_r = 1$):

$$\nabla^2 \mathcal{E} = \frac{\epsilon_r}{c^2} \frac{\partial^2 \mathcal{E}}{\partial t^2}. \quad (7.56)$$

On inserting the fields from eqn 7.50 and using eqn 7.52, this gives:

$$\begin{aligned} k_x^2 - (k_z^d)^2 &= \frac{\epsilon_d}{c^2} \omega^2, \\ k_x^2 - (k_z^m)^2 &= \frac{\epsilon_m}{c^2} \omega^2. \end{aligned} \quad (7.57)$$

Then, on using eqn 7.55 to eliminate the k_z components, we finally obtain:

$$k_x = \frac{\omega}{c} \left(\frac{\epsilon_m \epsilon_d}{\epsilon_m + \epsilon_d} \right)^{1/2} = \frac{\omega \sqrt{\epsilon_d}}{c} \left(\frac{\epsilon_m}{\epsilon_m + \epsilon_d} \right)^{1/2}. \quad (7.58)$$

This equation gives the dispersion curve for the surface plasmon polaritons.

Figure 7.16 compares the dispersion of the surface plasmon polaritons to the photon dispersion in the bulk of a metal with a dielectric constant given by eqn 7.7, namely:

$$\epsilon_m = 1 - \frac{\omega_p^2}{\omega^2}. \quad (7.59)$$

The relative permittivity of the dielectric is assumed to be real and independent of frequency.

Consider first the dispersion in the bulk of the metal, which is shown in Fig. 7.16(a). For frequencies below ω_p , the light is reflected, and there are

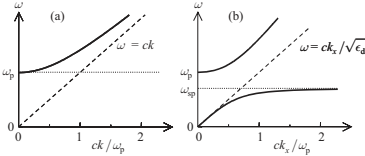


Fig. 7.16 (a) Photon dispersion in the bulk of a metal with a dielectric function given by eqn 7.7. (b) Surface plasmon polariton dispersion for the same metal. The curve is drawn for the case where the dielectric is air (i.e. $\epsilon_d = 1$). The dashed line in (b) shows the dispersion for light in the dielectric.

just evanescent fields in the medium. There are therefore no propagating modes with $\omega < \omega_p$. For $\omega > \omega_p$, the dispersion is given by (see eqn 7.45):

$$\omega = (\omega_p^2 + c^2 k^2)^{1/2}. \quad (7.60)$$

This asymptotes to $\omega = ck$ at large wave vectors.

The situation for the surface plasmon polaritons is qualitatively different, as shown in Fig. 7.16(b). The 'light line' for the dielectric defined by $\omega = ck_x/\sqrt{\epsilon_d}$ is shown for comparison. Three distinct frequency regions can be identified.

- (1) $0 < \omega < \omega_p/\sqrt{1+\epsilon_d}$. In this frequency region, both ϵ_m and $(\epsilon_m + \epsilon_d)$ are negative, so that k_x is real. For small ω , $|\epsilon_m|$ is large. Therefore the plasmon dispersion curve approaches the light line for small k_x .
- (2) $\omega_p/\sqrt{1+\epsilon_d} < \omega < \omega_p$. In this region ϵ_m is negative, but $(\epsilon_m + \epsilon_d)$ is positive. k_x is therefore imaginary, and there are no propagating modes.
- (3) $\omega > \omega_p$. Both ϵ_m and $(\epsilon_m + \epsilon_d)$ are now positive, so that real solutions for k_x are again found. At high frequencies, $\epsilon_m \rightarrow 1$, and the dispersion approaches the limit with $\omega = ck_x\sqrt{1+\epsilon_d}/\sqrt{\epsilon_d}$.

In region (1) at large k_x the group velocity (i.e. $d\omega/dk$) is zero and $\omega \rightarrow \omega_{sp}$. This asymptotic frequency limit is called the surface plasmon frequency. Equation 7.58 shows us that $k_x \rightarrow \infty$ when $(\epsilon_m + \epsilon_d) \rightarrow 0$, and so we can find ω_{sp} by solving $\epsilon_m(\omega) = -\epsilon_d$. For an undamped plasma with $\epsilon_m(\omega)$ given by eqn 7.59, we find:

$$\omega_{sp} = \frac{\omega_p}{\sqrt{1+\epsilon_d}}. \quad (7.61)$$

Note that $\omega_{sp} = \omega_p/\sqrt{2}$ when the dielectric is air.

The behaviour of the surface plasmon polaritons in region (1) where $\omega < \omega_{sp}$ is the most interesting, since this corresponds to propagating modes for frequencies below the plasma frequency. The spatial extent of the fields in the z direction can be found from the values of k_z in the

In electron energy-loss experiments on metals, it is common to observe peaks corresponding to both bulk and surface plasmons. This gives a convenient method for measuring both ω_p and ω_{sp} .

metal and the dielectric. This can be done by substituting eqn 7.58 back into 7.57 to obtain:

$$\begin{aligned}k_z^d &= \frac{\omega}{c} \left(\frac{-\epsilon_d^2}{\epsilon_m + \epsilon_d} \right)^{1/2}, \\k_z^m &= \frac{\omega}{c} \left(\frac{-\epsilon_m^2}{\epsilon_m + \epsilon_d} \right)^{1/2}.\end{aligned}\tag{7.62}$$

Since $(\epsilon_m + \epsilon_d)$ goes to zero at ω_{sp} , the decay constants increase with ω and diverge on approaching ω_{sp} . The field decay length l_z is equal to $1/|k_z|$, and this implies that the polaritons become more localized as ω approaches ω_{sp} . Note that it is usually the case that $|k_z^m| > |k_z^d|$, and hence that the plasmon extends further in the dielectric than in the metal, as illustrated in Fig. 7.15. At optical frequencies, l_z^d and l_z^m are typically a few hundred or few tens of nanometres, respectively. (See Exercise 7.20.)

Two other branches of plasmonics deal with the enhancement of radiative efficiencies and the increased transmission of light through sub-wavelength apertures. In the first case, the radiative efficiency of an atom in a dielectric can be enhanced by placing it close to a metal surface, and hence exploiting the large electric field amplitude that is present at the interface. In the second case, it has been demonstrated that plasmonic effects can enhance the transmission of light through periodic arrays of sub-wavelength-size holes in an optically thick metallic film. Details of these and other applications may be found in the Further Reading.

One of the aims of the research field of plasmonics is to propagate electromagnetic waves in a metal as surface plasmon polaritons. With values of l_z^m being in the sub-100 nm range, the waves can be confined to dimensions smaller than the wavelength of light, and we then enter the realm of **nanophotonics**, which is not accessible to conventional optics due to diffraction limits. The distance that the plasmon modes can propagate along the surface is determined by the imaginary part of ϵ_m . This can be several tens of micrometres (see Exercise 7.20), which is more than adequate for the applications that are being considered.

An issue that has to be addressed in plasmonics is the way to couple light to the polaritons. It is apparent from Fig. 7.16(b) that the polariton modes always lie below the light line. This means that polaritons can never transform directly into light: they are non-radiative modes. By the same token, it is not possible to couple light directly from the dielectric into the polariton modes, since it is never possible to match the wave vectors. Therefore, in order to couple external light into the surface plasmon polaritons, techniques must be used to change the wave vector of the light. One way to do this is to use a grating. In fact, it has been known for a long time that the reflectivity of metallic ruled gratings can drop significantly when one of the diffracted orders propagates parallel to the surface. This effect, which is called **Wood's anomaly**, is now known to be caused by the excitation of surface plasmon polaritons in the metal. Details of how the coupling to polariton modes is achieved in practice may be found in the references cited for Further Reading.

A striking example of surface plasmon effects is to be found in considering the optical properties of metal colloids. Metal colloids are made by dispersing a large number of very small metallic particles (i.e. 'nanoparticles') throughout a homogeneous medium such as glass or water. It has been known from antiquity that the colour of metal colloids is different from that of the bulk metal, and this phenomenon is now known to be caused by the resonant excitation of surface plasmons in the metal nanoparticles. However, in contrast to surface plasmon polaritons, the surface fields in the metallic nanoparticles do not propagate, since they

Colloids can be solids, liquids, or gases. The particle size in a colloid should be smaller than the wavelength of light. An interesting application of metal colloids is in the production of stained glass. The colouration of medieval stained glass is typically caused by gold, silver or copper nanoparticles incorporated into the glass during the melt process.

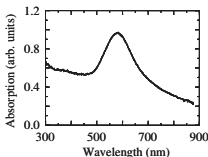


Fig. 7.17 Absorption spectrum for a thin film of gold nanoparticles embedded within an organic dielectric with $\epsilon_d \approx 2.5$. The film was grown on a glass substrate, and the spectrum was measured at room temperature. The particle size was 6–7 nm. Unpublished data from M.R. Sugden and T.R. Richardson.

are confined to the nanoparticles. They are therefore known as **localized surface plasmons**.

A simple explanation for the change of the optical properties of colloidal metals compared to the bulk can be given in terms of the polarizability. If the particle radius a is much smaller than the wavelength of the light, then it can be shown that the polarizability is given by:

$$\alpha = 4\pi a^3 \frac{\epsilon_m - \epsilon_d}{\epsilon_m + 2\epsilon_d}, \quad (7.63)$$

which has a resonance when $\epsilon_m = -2\epsilon_d$. In an undamped plasma, the resonance occurs at $\omega_p/\sqrt{3}$ when the dielectric is air (see Exercise 7.21), and is independent of the particle size. However, the resonance in real metals is shifted by interband absorption, and does depend somewhat on the size. Figure 7.17 shows the absorption spectrum of a thin film of gold nanoparticles embedded within an organic dielectric with $\epsilon_d \approx 2.5$. A strong plasmonic absorption peak centred at 580 nm (2.1 eV) is clearly resolved in the data. When the nanoparticles are suspended in water, the resonance occurs at higher frequencies owing to the lower permittivity of the dielectric. (See Exercise 7.22.) The absorption resonance is in fact in the green spectral region, and the colloid appears red, instead of the usual golden colour of the bulk metal. Similar effects can be observed for other metals.

The derivation of eqn 7.63, which has a similar form to the Clausius–Mossotti relationship given in eqn 2.35, may be found, for example, in Maier (2007). The general treatment of the interaction of light with conducting metal spheres is called **Mie theory**: see Born and Wolf (1999).

7.6 Negative refraction

Throughout this chapter we have been studying the optical properties of materials that have negative values of ϵ_r . We now wish to consider briefly the properties of materials that also have negative values of the relative magnetic permeability, μ_r . As we shall see, this possibility leads to the striking concept of a *negative* refractive index, which is a subject that has attracted much interest in recent years.

The general relationship between the refractive index of a medium and μ_r follows directly from Maxwell's equations. Equation A.29 shows that the speed of light in a medium is equal to $c/\sqrt{\epsilon_r\mu_r}$, and so we can write:

$$\tilde{n} = \sqrt{\epsilon_r\mu_r}. \quad (7.64)$$

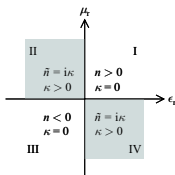


Fig. 7.18 Real and imaginary parts of the complex refractive index for four different combinations of values of ϵ_r and μ_r . The negative refractive index regime occurs in quadrant III, where both ϵ_r and μ_r are negative.

In everything that we have been considering so far in this book, we have been assuming that $\mu_r = 1$, so that eqn 7.64 reduces to $\tilde{n} = \sqrt{\epsilon_r}$. In fact, $(\mu_r - 1)$ characterizes the magnetic response of the medium, and magnetic dipoles respond to oscillating electromagnetic fields in much the same way that electric dipoles do. However, the natural resonant frequencies are low ($\sim$ GHz at most), and so the magnetic response is usually negligible at optical frequencies. It is therefore important to clarify at the start that there are no known natural materials that have μ_r significantly different from unity at optical frequencies. The materials that we shall be considering here are therefore purely artificial.

The four possible general combinations of values of ϵ_r and μ_r are depicted schematically in Fig. 7.18. In quadrant I, both ϵ_r and μ_r are positive. This is the usual situation in a transparent optical medium, and the solutions to Maxwell's equations are travelling waves with a phase velocity determined by a refractive index n that is both real and positive. In quadrant II, μ_r is positive, but ϵ_r is negative. This is the scenario that we have considered at length in this chapter, since it applies to the case of a metal below its plasma frequency. In these conditions, $\tilde{n}$ is purely imaginary. This means that the waves decay exponentially in the medium: there are no propagating solutions, and incoming waves from the air are reflected. A similar situation would occur in quadrant IV, where ϵ_r is positive, but μ_r is negative. The final case to consider is that which corresponds to quadrant III in which both ϵ_r and μ_r are negative. The properties of materials that fall into quadrant III were first considered theoretically by Veselago in 1968.

Veselago's main conclusion was that a medium with both ϵ_r and μ_r negative would be transparent, and have a *negative* refractive index. Not surprisingly, this gives rise to many unusual properties. The $\mathbf{k}$ vector of the wave and the direction of energy flow are in opposite directions. Since the energy flow is determined by the Poynting vector $\mathbf{E} \times \mathbf{H}$, the reversal of the direction of energy flow relative to $\mathbf{k}$ is consistent with reversing the direction of the magnetic field. This means that $\mathbf{E}$, $\mathbf{H}$ and $\mathbf{k}$ now form a left-handed system instead of the usual right-handed arrangement, and so materials with $n < 0$ are sometimes called 'left handed'.

Even more striking effects occur when light is refracted on entering the negative index medium. If the incoming ray has an angle of incidence θ_i , then the angle of refraction θ_r is given by Snell's law:

$$\frac{\sin \theta_i}{\sin \theta_r} = n. \quad (7.65)$$

When n is negative, θ_i and θ_r have opposite signs, as shown in Fig. 7.19(a). This leads to the possibility that the medium can behave as a lens, as shown in Fig. 7.19(b). An object to the left of the medium is brought to a focus to the right, after passing through an intermediate focus. The trajectory of the rays only depends on the thickness of the medium, and there are therefore no aberrations. Hence the medium is said to behave as a *perfect* lens. Since the rays emerge exactly as they would from the

source, the medium is invisible.

Interesting though these properties might be, there would be no point considering them if it were not possible to obtain a medium that has $n < 0$. Metals such as silver or gold have negative values of ϵ_r at most frequencies, and so the issue becomes that of obtaining the negative value of μ_r . As mentioned above, there are no known natural materials that possess this property at useful frequencies. Hence the negative refractive index must be engineered by creating an artificial structure called a **metamaterial**. It was not until the mid-1990s that this subject became of widespread interest, following work by Pendry in which the first practical designs for metamaterials with a negative refractive index were proposed.

The principle behind a metamaterial is to create a metallic structure designed to behave like a magnetic dipole resonator. Figure 7.19(c) illustrates one of the standard designs considered in the literature, namely an array of split rings. In this case the magnetic response is determined by the design of the constituent units (i.e. the split rings), which must be smaller than the wavelength of the electromagnetic waves, but are still much larger than the underlying atoms. It is for this reason that the medium is called a *metamaterial*.

We have seen in Chapter 2 that the refractive response of an electric-dipole resonator is negative above the natural frequency ω_0 . (See, for example, Fig. 2.6.) Magnetic resonators behave in a similar way, and so we can expect to obtain $\mu_r < 0$ in the frequency region above their resonant frequency. The difficulty is that the value of ω_0 depends on the size of the structure. A split-ring array with a period of ~ 10 mm has negative refraction at GHz frequencies, but much smaller structures must be used for optical frequency experiments. For this reason, the underlying principles of negative index materials are usually tested first at microwave frequencies. Further details of the design of metamaterials and progress in obtaining negative refraction at optical frequencies may be found in the works cited for Further Reading.

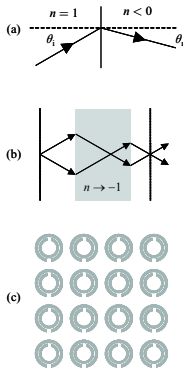


Fig. 7.19 (a) Negative refraction in a medium with $n < 0$. (b) Perfect lensing for $n < 0$. (c) Split ring design for producing a negative refractive index. In (a) and (b) the arrows give the direction of the Poynting vector.

Chapter summary

- Free electron effects are observed in metals and doped semiconductors. They can be modelled by the classical dipole oscillator model with no restoring force term. This approach is called the Drude–Lorentz model.
- The free electron plasma reflects strongly up to the plasma frequency, which depends on the electron density. The damping rate of the oscillations is determined by the momentum scattering time deduced from electrical conductivity measurements.

- Metals reflect strongly due to the plasma reflectivity effect. At frequencies above the plasma frequency, the metals become transparent. This effect is called the ultraviolet transparency of metals.
- Interband transitions are possible in metals from states below the Fermi energy to empty levels above it. The interband absorption can reduce the reflectivity from the value predicted by the Drude-Lorentz model, and must therefore be considered to obtain a good fit to experimental reflectivity data.
- Doped semiconductors reflect at frequencies in the infrared due to the free electrons and holes generated by the doping process. Free carrier absorption can be observed at frequencies above the plasma frequency but below the fundamental absorption edge at the band gap.
- P-type semiconductors show an additional absorption mechanism in the infrared due to intervalence band transitions.
- Doped semiconductors show sharp infrared absorption lines due to impurity transitions at low temperatures. At room temperature, the impurity states broaden the fundamental absorption edge.
- Plasma oscillations can occur at the plasma frequency. The quantized oscillations are called plasmons. These can be observed by electron energy-loss spectroscopy in metals, or by Raman scattering in doped semiconductors.
- Surface plasmons correspond to localized electromagnetic fields at the interface between a metal and a dielectric. Propagating surface plasmon polariton modes can be excited by coupling light to the metal with a grating or prism.
- Materials in which both the relative permittivity and magnetic permeability are negative are characterized by a negative refractive index. The artificial structures that show these effects are called metamaterials.

Further reading

The properties of electromagnetic waves in a conducting medium are covered in many electromagnetism and optics textbooks, for example Bleaney and Bleaney (1976), Born and Wolf (1999), or Hecht (2001).

The free carrier model of metals is covered in Singleton (2001). It is also covered in Ashcroft and Mermin (1976), Burns (1985), or Kittel (2005).

Free carrier reflectivity and absorption in semiconductors has been reviewed by Pidgeon (1980), and is also covered by Yu and Cardona (1996). Yu and Cardona give further details about intervalence band and impurity ab-

sorption.

The properties of plasmons are treated in depth in Maier (2007). Bulk plasmons are covered in Kittel (2005), while the classic text on surface plasmons is Raether (1988). Review articles on the research fields of plasmonics and nanophotonics may be found in Barnes et al. (2003), Lal et al. (2007), Maier and Atwater (2005), Murray and Barnes (2007), or Ebbeson et al. (2008).

The concept of negative refraction was proposed in Veselago (1968). Introductory reviews on the subject may be found in Pendry (2004) or Pendry & Smith (2004).

A more detailed review is given in Ramakrishna (2005). Progress on obtaining negative refraction at optical fre-

quencies is reviewed in Shalaev (2007).

Exercises

- (7.1) Derive a relationship between the Fermi energy E_F of a metal and its plasma frequency ω_p .
- (7.2) The ionosphere reflects radio waves with frequencies up to about 3 MHz, but transmits waves with higher frequencies. Estimate the free electron density in the ionosphere.
- (7.3) Estimate the skin depth of radio waves of frequency 200 kHz in sea water, which has an average electrical conductivity of about $4 \Omega^{-1} \text{m}^{-1}$. Hence discuss the difficulties you might encounter when attempting to communicate with a submerged submarine using radio waves.
- (7.4) Cesium metal is found to be transparent to electromagnetic radiation of wavelengths below 440 nm. Calculate a value for the electron effective mass using the data given in Table 7.1.
- (7.5) The momentum scattering time of silver is $4.0 \times 10^{-14} \text{ s}$ at room temperature. Calculate the dielectric constant at 500 nm, neglecting interband absorption effects. Hence estimate the reflectivity of a silver mirror at this wavelength. See Table 7.1 for the plasma frequency of silver.
- (7.6) Estimate the fraction of light with wavelength $1 \mu\text{m}$ that is transmitted through a 20 nm thick gold film at 77 K, where the DC electrical conductivity is $2 \times 10^8 \Omega^{-1} \text{m}^{-1}$. The plasma frequency and electron density of gold are given in Table 7.1.
- (7.7) Figure 7.20 shows the measured reflectivity of gold in the wavelength range 100–1000 nm. Account qualitatively for the shape of the spectrum, and deduce the energy gap between the d bands and the Fermi energy. Use the data to explain the characteristic colour of gold.
- (7.8) What is the value of the dielectric constant of a medium that has zero reflectivity? Use eqn 7.22 to show that the reflectivity of a lightly damped doped semiconductor is zero at the angular frequency given in eqn 7.25.
- (7.9) Use the data shown in Fig. 7.7 to deduce the value of the electron effective mass of InSb at each carrier density. Take $\epsilon_{\text{opt}} = 15.6$.
- (7.10) The absorption coefficient at room temperature of an n-type sample of InAs with a doping level of $1.4 \times 10^{23} \text{ m}^{-3}$ is found to be 500 m^{-1} at $10 \mu\text{m}$. Estimate the momentum scattering time, given that the electron effective mass is $0.023 m_0$ and the refractive index is 3.5.

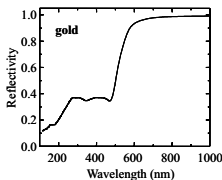


Fig. 7.20 Reflectivity of gold in the wavelength range 100–1000 nm. Data from Lide (1996).

- (7.11)* A laser beam operating at 632.8 nm with an intensity of 10^6 W m^{-2} is incident on a sample of pure InP at room temperature. The absorption coefficient at this wavelength is $6 \times 10^6 \text{ m}^{-1}$, and the carrier lifetime is 1 ns. Estimate the free carrier absorption coefficient at the wavelength of a CO_2 laser ($10.6 \mu\text{m}$), where the refractive index is 3.1. The effective mass and momentum scattering time for the electrons are $0.08 m_0$ and $2 \times 10^{-13} \text{ s}$, while the equivalent values for the holes are $0.6 m_0$ and $5 \times 10^{-14} \text{ s}$.
- (7.12)* Consider the intervalence band processes illustrated in Fig. 7.9 for a heavily doped p-type sample of GaAs containing $1 \times 10^{26} \text{ m}^{-3}$ acceptors.

*Exercises marked with an asterisk are more difficult.

The valence band parameters for GaAs are given in Table D.2.

- (a) Work out the Fermi energy in the valence band on the assumption that the holes are degenerate. What are the wave vectors of the heavy and light holes at the Fermi energy?

(b) Calculate the upper and lower limits of the photon energies for the three absorption processes labelled (1), (2), and (3) in Fig. 7.9, namely the $lh \rightarrow hh$, the $SO \rightarrow lh$ and the $SO \rightarrow hh$ transitions.

- (7.13) Figure 7.11 shows the infrared absorption spectrum of n-type silicon, which has a dielectric constant of 16. Two series of lines labelled np_0 and $np_{\pm}$ are identified in the data.

(a) Show that the np_0 series is consistent with eqn 7.30, and deduce a value for the electron effective mass for these transitions.

(b) Show that the $np_{\pm}$ series follows the following formula:

$$h\nu = \frac{R_0^*}{12} - \frac{R_{\pm}^*}{n^2},$$

stating the values of R_0^* and $R_{\pm}^*$ deduced from the data.

- (7.14) It is found that the infrared absorption spectrum of a lightly doped n-type semiconductor with a relative permittivity of 15.2 consists of a series of sharp lines at low temperatures. The energies of the lines are given by:

$$E(n) = R^* (1 - 1/n^2)$$

where R^* is 2.1 meV and n is an integer greater than 1. Explain why the energies of the absorption lines are almost independent of the type of impurity atoms used for the doping, and deduce a value for the electron effective mass.

- (7.15) The fundamental absorption edge of a semiconductor shifts from 5.26 μm to 5.44 μm when doped with acceptors. Deduce a value for the ground state acceptor level energy relative to the valence band.

- (7.16) The beam from an argon ion laser operating at 514.5 nm is incident on an n-type GaAs sample. A peak is observed in the intensity of the scattered light at 534.3 nm. Explain this observation, and estimate the electron density, given that $m_e^* = 0.067 m_0$ and $n = 3.3$.

- (7.17) Calculate the doping density at which the plasmons in n-type GaAs have the same wave number as the longitudinal optic phonon at 297 cm^{-1} . Take $m_e^* = 0.067 m_0$ and $n = 3.3$.

- (7.18) In a metal, the frequency at which $\epsilon_r = 0$ varies slightly with the wave vector $k \equiv |\mathbf{k}|$, and for small k we have:

$$\omega(k) \approx \omega_p \left(1 + \frac{3v_F^2 k^2}{10\omega_p^2} \right),$$

where v_F is the Fermi velocity of the electrons. Consider a metal with electron density $N = 10^{29} \text{ m}^{-3}$ and lattice constant $a = 4 \text{ \AA}$. Calculate the relative size of the term in k^2 for $k = 0.1 \pi/a$, i.e. 10% of the size of the Brillouin zone.

- (7.19) In an electron energy-loss experiment on a metal, two series of peaks are observed that obey eqn 7.48 with $\hbar\omega_p = 10.3 \text{ eV}$ and 15.3 eV . Use the data in Table 7.1 to determine which metal is being investigated, and account for the two series of peaks.

- (7.20) A surface plasmon mode with a frequency corresponding to a vacuum wavelength of 600 nm is excited at the interface between silver and air. The relative permittivity of silver at this frequency is given approximately by $\epsilon_m = -18 + i$.

(a) Calculate the decay constants k_z^a and k_z^m , and hence deduce the field decay lengths in the direction normal to the surface in both the air and in the metal.

(b) The propagation length in the direction parallel to the surface is defined as the distance over which the intensity drops by a factor of e^{-1} . Calculate the imaginary part of k_z , and hence deduce the propagation length for silver at 600 nm.

- (7.21) Show that the resonance of the polarizability of colloidal metal nanoparticles that obey eqn 7.7 occurs at an angular frequency given by $\omega_p/\sqrt{1+2n^2}$ in a medium with a refractive index of n . Evaluate this frequency for the case where the dielectric is air.

Table 7.3 Complex relative permittivity of gold between 500 and 550 nm. Adapted from Raether (1988).

Wavelength (nm)	ϵ_1	ϵ_2
500	-2.3	3.6
510	-3.0	3.1
520	-3.7	2.7
530	-4.4	2.4
540	-5.2	2.2
550	-6.0	2.0

- (7.22) (a) The complex relative permittivity of gold in the range 500–550 nm is given in Table 7.3. Use this information to estimate the resonance wavelength of colloidal gold nanoparticles in water, which has

a refractive index of 1.33.

(b) Use the data for gold given in Table 7.1 to compare the result obtained in part (a) with the pre-

diction of the previous exercise. Account for any difference.